City Signals: What 311 Service Requests Reveal About Community Needs

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Objective

The objective of this project is to analyze 311 City Service Requests to understand the types of issues residents report most often, where requests are concentrated, when requests are submitted, and how long it takes for requests to be resolved. This project uses data cleaning, exploratory data analysis, and visualization to identify patterns in city service needs.

Introduction

The dataset selected for this project is the 311 City Service Requests dataset from DC Open Data. This dataset contains records of service requests submitted by residents to report city-related concerns in Washington, DC. These requests include public service issues such as bulk collection, parking enforcement, missed trash collection, missed recycling collection, sanitation enforcement, and other concerns that affect everyday life in the city.

This dataset is interesting because it turns ordinary residents' concerns into measurable information. A 311 request may look small when viewed by itself, but thousands of requests together can reveal patterns about what residents experience, what services they need, and where city resources are under the most pressure. For this project, I worked with 440,600 service request records and 39 cleaned columns. The dataset included information such as request type, request status, ward, date submitted, resolution date, priority, agency, and response time.

The main purpose of this analysis is to understand how residents use 311 services and what those requests reveal about community needs. I focused on five questions: what types of requests are most common, which wards have the most requests, when requests are most often submitted, how long requests usually take to close, and which request types take the longest to resolve. These questions helped guide the cleaning, analysis, visualizations, and interpretation in the project.

Method

The analysis began with data understanding. I loaded the dataset into Python using pandas and reviewed the first rows, number of rows and columns, column names, data types, and summary statistics. This step helped me understand which columns were most useful for the project. Important variables included servicecodedescription for the request type, serviceorderstatus for the request status, ward for location, adddate for the date submitted, resolutiondate for the date closed, and response_time_days for the time it took to resolve a request.

The next step was data cleaning. I standardized all column names by making them lowercase and replacing spaces and special characters with underscores. This made the dataset easier to use in Python. I also checked for missing values. Several columns, including inspectiondate, created, edited, inspectionflag, status_code, and inspectorname, were completely empty and therefore not useful for this analysis. The details column also lacked 99.82 percent of its values, so it was not

considered a major analytical feature. Some useful columns had smaller amounts of missing data, such as resolutiondate and response_time_days, which were missing 1.39 percent of values.

I also checked for duplicate rows. The dataset had 0 duplicate rows, so no records were removed for duplication. I converted the adddate and resolutiondate columns into a datetime format so they could be used for time-based analysis. Then I created new columns for year, month, month name, day of the week, and hour. These new columns enabled analysis of when service requests were submitted. Response time was calculated in days by subtracting the add date from the resolution date. If a response time was negative, it was treated as invalid because a request cannot logically be resolved before it was submitted.

Storytelling: Data Visualization and Interpretation

The visualizations in this project were created to move from broad patterns to more specific insights. I first looked at the most common request types, then examined request status, location, time patterns, and response time. Each graph was used to answer one part of the research question and to connect the data back to community needs.

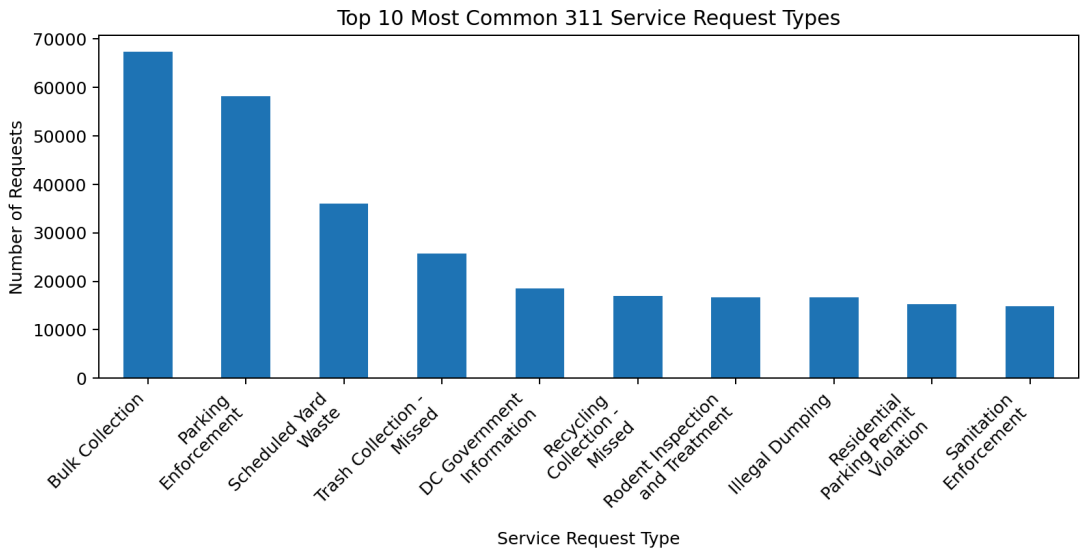

Figure 1. Top 10 most common 311 service request types.

The first visualization shows that Bulk Collection was the most common service request type, with 67,414 requests. Parking Enforcement was second with 58,149 requests, and Scheduled Yard Waste was third with 36,011 requests. This pattern shows that many of the most common 311 requests are connected to daily neighborhood maintenance, waste collection, and parking concerns. These are not abstract issues. They are the types of public service needs that residents notice as they move through their neighborhoods, maintain their homes, or use shared community spaces.

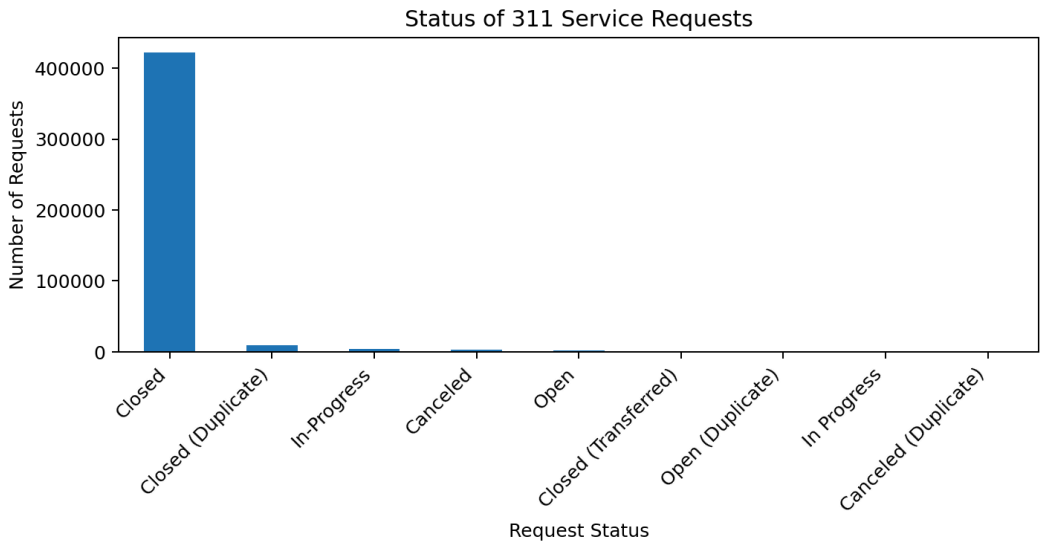

Figure 2. Status of 311 service requests.

The request status visualization shows that most records were marked as closed. Out of the 440,600 total requests, 422,406 were listed as Closed. This suggests that the majority of requests in the dataset eventually moved through the city service process. However, the data also included requests marked as Closed Duplicate, In Progress, Canceled, Open, and Closed Transferred. These smaller categories matter because they show that not every request follows the same path after submission.

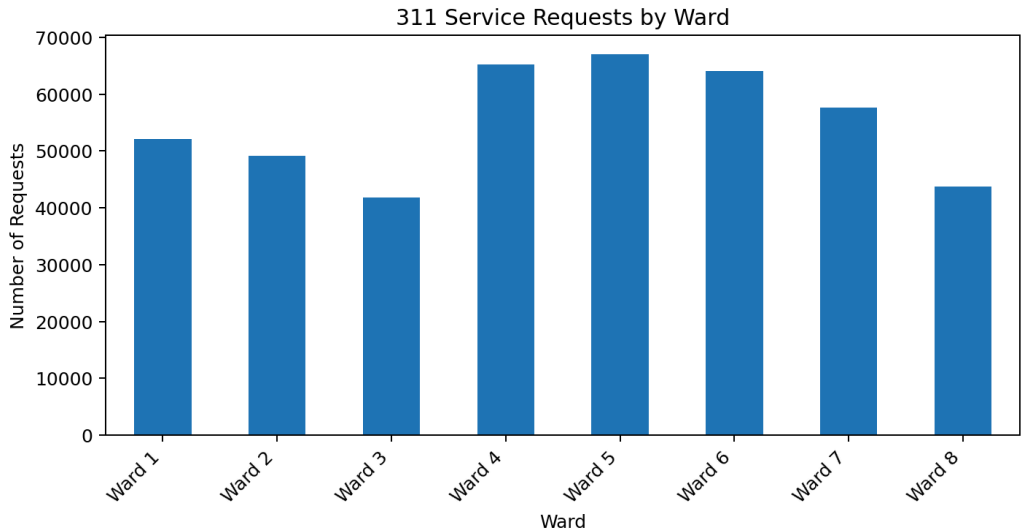

Figure 3. 311 service requests by ward.

The ward visualization shows that service requests were not evenly distributed across the city. Ward 5 had the highest number of requests with 66,950, followed by Ward 4 with 65,170 and Ward 6 with 64,108. Ward 3 had the fewest requests, with 41,816. These differences may reflect population size, infrastructure needs, reporting habits, or differences in the kinds of services residents need. The graph is important because it shows that city service demand has a spatial pattern, not just a general citywide pattern.

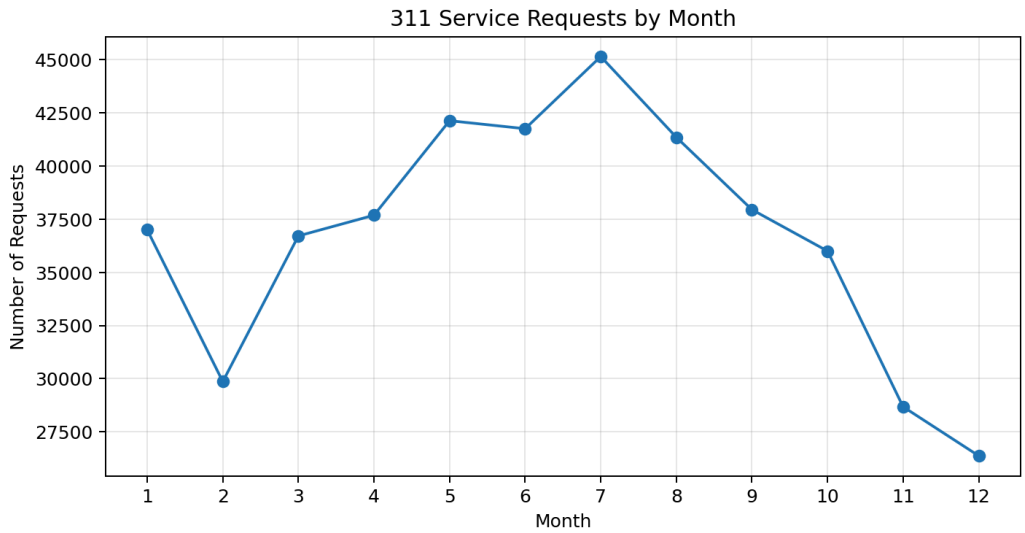

Figure 4. 311 service requests by month.

The monthly trend shows that July had the highest number of requests with 45,137. May, June, and August were also higher than the rest of the year. December had the fewest requests, with 26,378. This pattern suggests that 311 request volume may increase during warmer months. Seasonal factors may influence what residents notice and report, such as yard waste, sanitation concerns, outdoor maintenance, parking, and public space issues.

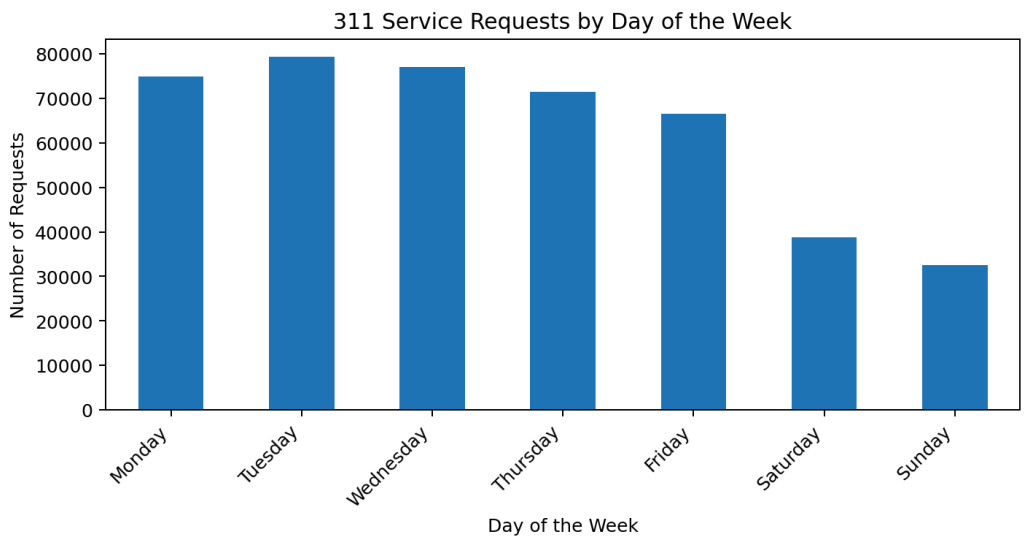

Figure 5. 311 service requests by day of the week.

The day-of-the-week chart shows that Tuesday had the highest number of requests, with 79,260, followed by Wednesday with 77,076 and Monday with 74,904. Saturday and Sunday had the lowest request totals. This suggests that residents are more likely to submit 311 requests during the workweek. One possible explanation is that residents may report issues when city offices and services feel more active or when they are following up on problems during regular weekday routines.

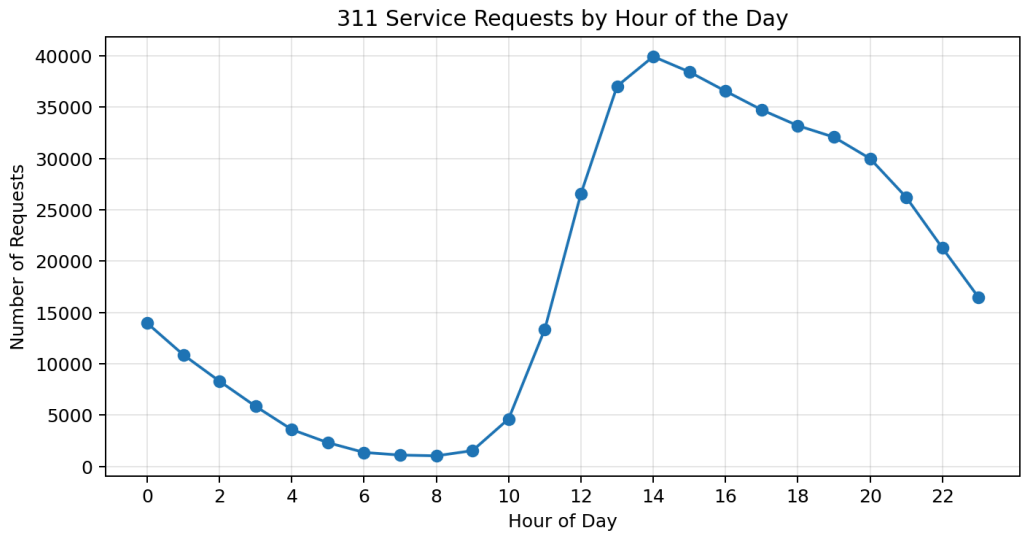

Figure 6. 311 service requests by hour of the day.

The hourly graph shows a clear increase in requests during the afternoon. The busiest hour was 2:00 PM, with 39,933 requests, followed closely by 3:00 PM and 1:00 PM. Request volume was lowest in the early morning hours. This pattern suggests that many residents submit requests during the middle of the day, when they may be active online, contacting city services, or noticing problems while moving through the city.

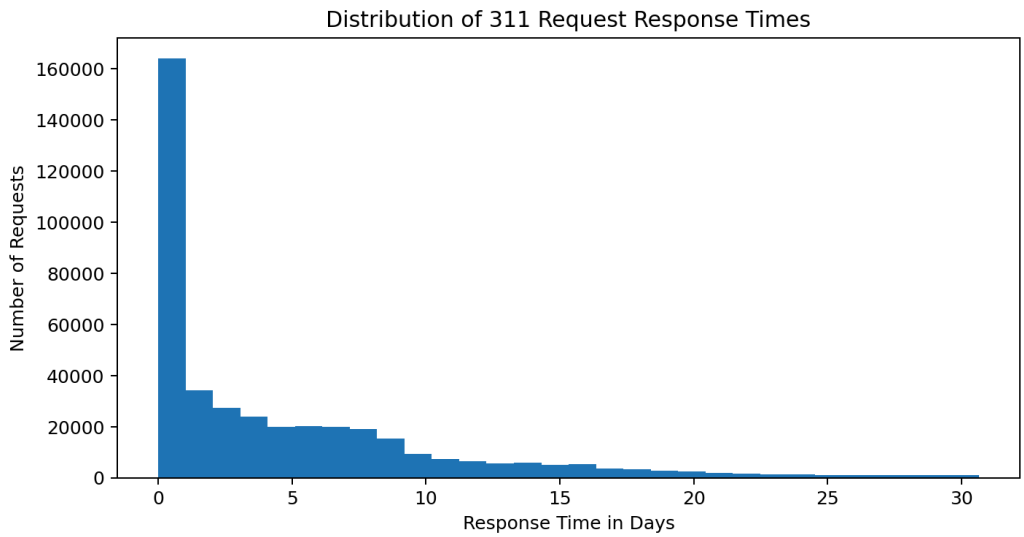

Figure 7. Distribution of response time in days.

The response time distribution shows that most requests were resolved within a relatively short time, but some requests took much longer. The average response time was 9.32 days, while the median response time was 2.88 days. The difference between the average and the median is important. It shows that a smaller number of requests with very long response times pulled the average upward. This means the typical request was resolved faster than the average alone might suggest.

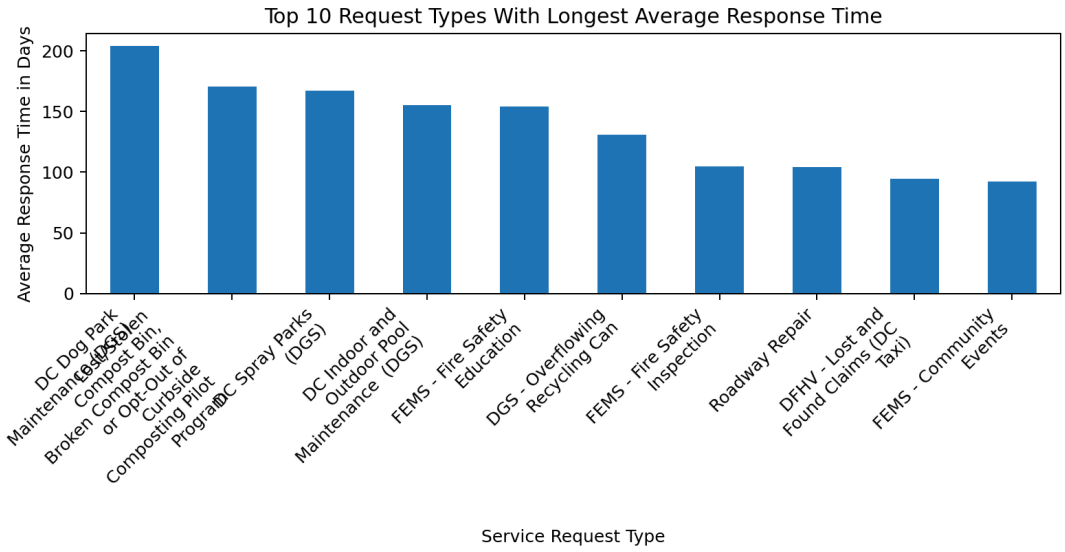

Figure 8. Request types with the longest average response time.

The final visualization looks at the request types with the longest average response times. DC Dog Park Maintenance had the longest average response time at about 204 days. Lost or stolen compost bins and curbside composting requests averaged about 171 days, while DC Spray Parks averaged about 167 days. These categories show that the slowest requests are not necessarily the most common. Some service issues may require more coordination, specialized staff, seasonal scheduling, or longer agency processes.

Conclusion

Overall, this project shows that 311 service request data can reveal important patterns about community needs and city service response. The analysis showed that the most common requests were connected to practical daily issues, especially bulk collection, parking enforcement, yard waste, trash collection, and recycling collection. These findings suggest that many residents use 311 to report issues that directly affect cleanliness, order, and access in their neighborhoods.

The data also showed that service requests varied by location and time. Ward 5 had the highest number of requests, while Ward 3 had the lowest. July was the busiest month, Tuesday was the busiest day of the week, and 2:00 PM was the busiest hour of the day. These patterns show that city service demand is not random. It changes by place and time, and those changes can help city agencies understand when and where residents are most likely to need support.

The response time analysis added another layer to the project. Most requests were closed, and the median response time was under three days, but some request types took much longer to resolve. This shows that a city can close many requests while still experiencing longer delays in certain categories. In a broader context, this matters because response time affects how residents experience city services. When requests take a long time to close, residents may feel that their concerns are not being addressed quickly, even if the request is eventually completed.

One limitation of this analysis is that the dataset only includes reported problems. It does not include issues that residents noticed but did not report. Another limitation is that response time may be affected by factors not fully visible in the dataset, such as staffing, weather, agency workload, the severity of the issue, or budget constraints. In the future, this analysis could be

expanded by comparing 311 requests with population, neighborhood, or city budget data. That would make it possible to better understand whether some communities experience higher service demand, longer response times, or different kinds of public service needs.

The most important takeaway is that 311 data is more than a record of complaints. It is a record of how residents experience the city in everyday life. Each request represents someone noticing a problem and asking the city to respond. When the requests are analyzed together, they become a story about community needs, service demand, and the relationship between residents and local government.

References

DC Open Data. 311 City Service Requests Dataset.

Pandas Development Team. pandas Python Data Analysis Library.

Matplotlib Development Team. Matplotlib Visualization Library.

Plotly Technologies Inc. Plotly Python Graphing Library.

Acknowledgments

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